

Generative Adversarial Network for Text-to-Face Synthesis and Manipulation

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Figure 1: Several experimental results on our proposed *FFHQ-Text* dataset (Left) and *Multi-Modal-CelebA-HQ* dataset (Right) [28] for facial image synthesis and manipulation (Add, Change and Delete) from input text descriptions. (Zoom in for better view.)

ABSTRACT

Over the past few years, several studies have been conducted on text-to-image synthesis techniques, which transfer input textual descriptions into realistic images. However, facial image synthesis and manipulation from input sentences have not been widely explored due to the lack of datasets. My research interests center around the development of multi-modality technology and facial image generation with Generative Adversarial Networks. Towards that end, we propose an approach for facial image generation and manipulation from text descriptions. We also introduce the first Text-to-Face synthesis dataset with large-scale facial attributes. In this extended abstract, we first present the existing condition and further direction of my Ph.D. research that I have followed

during the first year. Then, we introduce the proposed method (accepted by IEEE FG2021), annotated novel dataset and experimental results. Finally, the future outlook on other challenges, proposed dataset and expected impact are discussed. Codes and paper lists studied in text-to-image synthesis are summarized on <https://github.com/Yutong-Zhou-cv/Awesome-Text-to-Image>.

CCS CONCEPTS

• Information systems → Multimedia databases; • Computing methodologies → Computer vision; Image processing; Image representations.

KEYWORDS

Datasets; Text-to-image synthesis; Generative Adversarial Network; Vision and language

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1 MOTIVATION

With the development and promotion of multi-modal technology, a better combination and understanding between vision and language are becoming a promising candidate for future human-machine interaction. Recently, researchers are focusing on jointing visual information and semantic information in traditionally independent fields of Computer Vision (CV) and Natural Language Processing (NLP) on several comprehensive tasks, such as Vision-Language Navigation [25], Vision-Language Representation Learning [8], Video-Based Text Generation [12], *etc.*

However, when the content cannot be expressed directly by visual information, it is possible to produce some textual descriptions in advance and convert them into corresponding images to provide additional visual information. We called this Text-to-Image Synthesis, which has extensive application prospects image editing, image retrieval, *etc.* Three necessary conditions are expected to be considered for this challenging task: (1) Generate high-quality and high-resolution images. (2) Generate well-consistent and identifiable images with semantic content. (3) Semantically modify particular visual features according to the input text descriptions while other irrelevant features are maintained constant.

Nowadays, as one of the earliest practical applications of text-to-image technology in real life, visual reproduction of a suspect's appearance based on eyewitnesses' memory and verbal description still plays an important role in the criminal investigation field. Moreover, text-to-face synthesis also possesses unique potential in human retrieval, public security, missing child rendering, *etc.* However, the testimony provided by the witness may differ from the actual physical appearance caused by fear or other reasons. Therefore, manipulating particular visual features of the synthetic facial images corresponding to the input description, while retaining the other attributes becomes increasingly important in text-to-face synthesis technology for the public safety domain.

2 RELATED WORK

2.1 Text-to-image Synthesis

There are a variety of published approaches to text-to-image synthesis tasks. Most contributions of these papers rely on multi-model learning methods that exploit Generative Adversarial Network(GAN) and deep convolutional decoder networks [1]. In 2016, Reed *et al.* [18] first applied conditional Generative Adversarial Network(cGAN) to generate images from text descriptions. Then Reed *et al.* proposed another model called GAWWN [19] which is used to synthesize images. In 2017, Zhang *et al.* proposed two text-to-image synthesis methods called StackGAN [30] and StackGAN++ [31], which have the ability to generate images with multiple-stage from coarse to fine. In 2018, Xu *et al.* proposed AttnGAN [29] which implements a multi-stage module [31] by using attention mechanisms. In 2019, Qiao *et al.* proposed a novel global-local attention module and text-to-image-to-text network called MirrorGAN [16], which generates images from coarse to fine by simultaneously using both local word attention and global sentence attention. In 2020, different from previous works that only provided single-caption to synthesis images, Cheng *et al.* proposed a novel text-to-image synthesis called RiFeGAN [5], which exploit multi-caption Generative Adversarial Networks to synthesize images from several

features. In 2021, Ramesh *et al.* proposed a novel approach based on a transformer that constructs autoregressive models of text and image tokens as a single data stream, called DALL-E [17], which is a 12-billion parameter version of GPT-3 [2] trained for zero-shot text-to-image generation task.

In previous years, the basic approach of facial image generation is drawing sketches by artists, which can help eyewitnesses recall more details about the suspect's appearance step by step. This process is complex because all the descriptions of the suspect's appearance are recalled from the witness's memory. Moreover, to accurately match the sketches drawn with the descriptions of witnesses, artists also need numerous training in drawing face sketches. However, in many cases, there are only facial descriptions of suspects are available. Therefore, text-to-face synthesis is more challenging due to the ambiguity and complexity of text description and has more practical significance and application prospects.

Despite the major challenges mentioned above, researchers still raised some encouraging works committed to text-to-face synthesis. Nasir *et al.* [14] firstly generated captions for images automatically by exploiting the attribute labels of the CelebA [13] dataset and then explored the text-to-face generation in the same latent space. However, the generated images are only 64×64 pixels, which cannot recognize the details of the facial features. Recently, Wang *et al.* [26] proposed a text-to-face model, which can produce high-resolution images (1024×1024) by adopting the pre-trained model of StyleGAN2 [9].

However, the existing studies mainly focus on generating unique images from text descriptions, which can not effectively manipulate particular local features corresponding to the input text descriptions while maintaining global features and irrelevant attributes.

2.2 Text-to-image datasets

Table 1 shows a list of *Text-to-X* datasets that have been widely adopted in related works and *Text-to-Face* datasets for facial image generation task that proposed in recent years. The commonly used datasets for text-to-image synthesis research are CUB-200-2011 [24], Oxford-102 Flowers [15] and MS-COCO [11]. Compared with *Text-to-X* datasets, the existing *Text-to-Face* datasets have fewer image-text pairs and have not been widely released and used.

In our view, one of the most important factors significantly hinders the progress of automatic text-to-face synthesis & manipulation research is the lack of suitable datasets. The prime reason is that collecting and annotating facial images is a burdensome task. Moreover, facial features have less correlation with each other than bird or flower images. More generally, the descriptions of birds and flowers are mostly about the color of feathers and petals. Unlike the above features, facial descriptions can be much more complicated caused by ethnicity, gender, age, surroundings, and other features. The only public text-to-face dataset—Multi-Modal CelebA-HQ [28] which contains large-scale face images but only small-scale facial attribute annotations and simple text descriptions, which cannot generation complete facial images with diverse characteristics. To the best of our knowledge, there is no publicly available text-to-face synthesis dataset with large-scale facial attributes. Therefore, we collect and manually annotate a novel dataset with multi-attributes called **FFHQ-Text** (Details are shown in Section 4).

Table 1: Text-to-image datasets can be divided into *Text-to-X* (top three rows) and *Text-to-Face* (bottom four rows), which generate birds, flowers, iconic objects/scenes and facial images from input sentences respectively. "Annotation" denotes the number of text descriptions per image. "Attrs" denotes the total number of attributes contained in each dataset.

Dataset Name	Public	Content				
		Categories	Images (Resolution)	Annotations	Attrs	Other Details
CUB-200-2011 [24]	✓	200	11,788 (Unfixed)	10	Uncounted	BBox, Segmentation...
Oxford-102 Flowers [15]	✓	102	8,189 (Unfixed)	10	Uncounted	-
MS-COCO [11]	✓	91	120k (Unfixed)	5	Uncounted	BBox, Segmentation...
SCU-Text2face [4]	×	1(Mixed)	1,000 (256×256)	5	Unknown	-
Text2FaceGAN [14]	×	1(Mixed)	10,000 (178×218)	6	40	-
Multi-Modal CelebA-HQ [28]	✓	1 (Mixed)	30,000 (512×512)	10	38	Masks, Sketches
FFHQ-Text(Ours)	Coming soon	1 (Female)	760 (1024×1024)	9	162	BBox

3 APPROACH

Our main goal is to design an algorithm that can synthesize high-quality and realistic face images from the given text descriptions. Furthermore, when we modify text descriptions, the synthesized image should be well-matched with the semantic meaning of the modified text description while preserving not mentioned content.

In our work, we adopt the recent multi-stage architectures of text-to-image synthesis researches: AttnGAN [29] as our baseline architecture. There are four stages in our architecture, which contain four scales are 64×64 , 128×128 , 256×256 and 512×512 to synthesize high-resolution images from coarse to fine. This section discusses two parts of our approach: spatial and channel-wise attention and cyclic generative adversarial network with identity preserving loss function.

3.1 SCA: Spatial and Channel-wise Attention

Note that AttnGAN [29] is not perfect in capturing global coherent structures, which cause the synthesized images are not likely realistic and local details distortion. We aim to focus on global information to important features and suppressing unnecessary ones. To achieve this, we introduce a word-level spatial-wise and channel-wise mixed attention module that utilize the related features from relevant words and different visual portions to enhance more details. Our proposed attention model is shown in Fig 2.

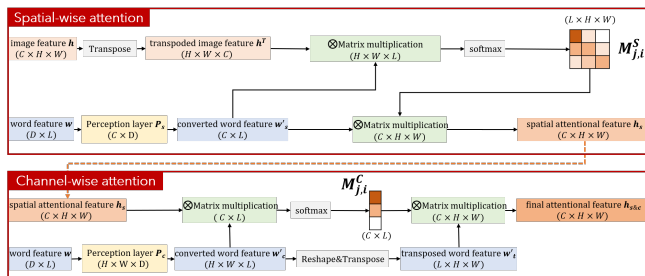


Figure 2: The architecture of proposed spatial and channel-wise attention modules.

Spatial-wise attention mainly focuses on 'where' is a spatial informative part [27], which extracts the image feature for recognizing individual words and convert them into a word of context vectors.

The spatial attention takes word features and image features from the previously hidden layer as input and outputs the word-context vector with spatial-wise attention to the semantic-related regions.

Channel-wise Attention mainly focuses on 'what' is important of the image features, which attempts to allocate different significance to the channels of feature maps. The channel-wise attention can be viewed as the process of selecting semantic attributes on the demand of the word-context [3], which associates semantically meaningful parts with meaning corresponding words. Therefore, the channel-wise attention takes these corresponding word-context features are further concatenated with the word features and regarded as input for image synthesizing at the next stage.

3.2 Cyclic Generative Adversarial Network

Inspired by CycleGAN [32] and MirrorGAN [16], We propose a cyclic generative adversarial network, which combines text-to-image synthesis and image captioning. For the image captioning task based on [23], we adopt a CNN that maps images to semantic vectors, which is built upon the Inception-v3 model [21] pretrained on ImageNet as an image encoder for extracting features of images. The target captions are placed to the decoder part of the pre-trained BERT model to obtain predictions for all sequence positions. Furthermore, we also propose an identity preserving loss function to maintain the global features while manipulating.

4 FFHQ-TEXT DATASET

As already mentioned in Section 2.2, there is only one publicly available text-to-face synthesis dataset with small-scale attributes and large-scale images named *Multi-Modal CelebA-HQ* [28]. Therefore, in order to support this research topic and fill in the gap of datasets, we annotate a novel text-to-face dataset named *FFHQ-Text*, which is the first text-to-face dataset with large-scale facial attributes.

4.1 Image Collection

In order to build a standard text-to-face dataset, we firstly download all facial images from Flickr-Faces-HQ Dataset (FFHQ) web page¹. Then, we manually ensure that the downloaded high-quality and high-resolution (1024×1024) facial images actually only contain one complete human face. Up to now, we select a total of 760 images of females (girl/woman) in the wild as one category.

¹<https://github.com/NVlabs/ffhq-dataset>

4.2 Annotation of Facial Features

Facial feature selection and description are crucial because of two factors: the reliability of facial features and the lack of accuracy in witnesses' descriptions [10]. In this study, we explore standardized terminology in the human facial to setup our dataset FFHQ-Text, which is breakdown into 13 multi-valued facial element groups (Facial orientation, Eyes, Eyebrows, Nose, Mouth, Ear, Skin, Face, Hair, Headwear, Accessory, Age, Gender) from local to global with 162 attributes. Furthermore, for each face image of different people with different facial attributes, 9 sentences are annotated by 3 different persons to maintain the great diversity in terms of patterns and species of our defined attributes.

Moreover, to determine the location and orientation of the face more accurately, bounding boxes were obtained by VGG Image Annotator [6], which is a simple and standalone manual annotation software² for image, audio and video.

In particular, for the definition of the "Age" element, Eran *et al.* [7] first raised that not all faces could be reliably marked with gender (mostly babies) and not all could be marked with age. Thus, we set 8 age categories that related with geometric and appearance throughout a person's whole life (0-2, 4-6, 8-13, 15-20, 25-32, 38-43, 48-53, over 60) the same as [7]. Furthermore, to create a sub-cluster in the female category, we define the first four age groups as "Girl" and the last four age groups as "Woman".

4.3 Future Extensions

We are still in the process of expanding the dataset and set an idealized target of 10,000 facial images (Male & Female) to be annotated with beyond 250 facial attributes. In the future, we will (1) expand male images from the FFHQ dataset and set different categories with the female group in order to generate facial images accurately; (2) explore public facial processing platforms to get more details of facial features, such as head pose angles, eyes occlusion scores, face sketches, *etc.*; (3) establish specific rules and pruned images with gender confidence, age confidence, and head orientation angles to generate and manipulate images more specifically.

In order to facilitate future research on text-to-face synthesis & manipulation research and other related studies (*e.g.*, text-guided image retrieval & editing, image-text matching, *etc.*), our FFHQ-Text dataset will be made publicly available to the research community.

5 RESULTS

To demonstrate the performance of our proposed method for text-to-image synthesis, we adopt the proposed FFHQ-Text and Multi-Modal CelebA-HQ [28] datasets for our experiments. As a result, our proposed method performs well both qualitatively (see three groups of examples in Fig. 1) and quantitatively (see IS score in Table. 2).

For quantitative evaluation, we adopt the Inception Score (IS) [20] for our evaluation metrics. This metric rewards high-quality and varied images and correlates best with human judgments. To make a comparison with AttnGAN [29], DF-GAN [22], we select 10 partitions of 500 randomly generated samples to compute the Inception Scores values shown in Table 2 on our FFHQ-Text dataset. Obviously, the proposed method is superior to the other two methods.

²<https://www.robots.ox.ac.uk/~vgg/software/via/>

Table 2: Quantitative evaluation for AttnGAN, DF-GAN and ours on FFHQ-Text dataset.

Methods	Inception Score $\uparrow$
AttnGAN [29]	2.27 $\pm$ 0.28
Ours (Section 3.1)	2.57 $\pm$ 0.22
DF-GAN [22]	2.66 $\pm$ 0.20
Ours (Section 3.1+3.2)	3.05$\pm$0.26

6 IMPLICATIONS AND FUTURE OUTLOOK

This section discusses ethical considerations, future direction and expected impacts in a wide range of text-to-image fields.

Ethical & privacy Implications: Nowadays, facial recognition technology provides public safety and the judicial system an effective way out of cumbersome and time-consuming processes for catching criminals accurately and offering evidence for convictions. However, the question of ethics and privacy is the most contentious one. Our research is designed to generate facial images with an approximation of a photorealistic human appearance from text descriptions, which mainly apply to arts and entertainment purposes (makeup design and transfer, Virtual Reality photography manipulation, *etc.*). However, when this technology is adopted for other critical or commercial purposes (missing child rendering, criminal investigation, *etc.*), fake facial images generated from GAN have an inherent bias and might cause inaccurate. Therefore, professional judges should critically analyze the synthesized images and develop a regulatory framework to prevent ethical abuses.

Future Research: This paper is a part of my research thesis. During the first year of my doctoral studies, I studied state-of-the-art text-to-image generation technology, comprehended correlations from multi-modal information processing and proposed a novel approach on text-to-face synthesis & manipulation. To conduct a novel text-to-face dataset with large-scale attributes, I investigate human face terminology, which further works on text-to-face synthesis and supports manipulate diversified facial attributes.

For the remainder of my doctoral studies, I consider adopting multi-modality based image generation and manipulation, combining other fields of human perception (visual sentiment, speech, even olfactory sensation) to contribute to the multimedia field and social work. Furthermore, I aim to communicate with researchers in dissimilar fields (mathematics, physics, humanities, psychology, *etc.*), integrate the best of these technologies, and conduct researches that can be well applied to a wide range of fields.

Expected Impacts: For manufacturing industry design applications, it is capable of automatically generating computer sketches and combining the general-purpose modeling technique alongside large-scale data for the Computer-Aided Design (CAD), Fashion Design, *etc.* For commercial applications, this technology can be used as an auxiliary tool for video conferences to provide participants with a real-time image-aided understanding based on the speaker's descriptions and human-made PowerPoint.

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